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1. How This Project Makes the World Different

The current paradigm of clinical documentation relies on ambient scribing tools that produce fluent, generative prose. While these tools summarize dictations well, they introduce a dangerous new friction into the Electronic Health Record (EHR) workflow: hallucination verification.

Consider our primary persona, Dr. Aisha, a general practitioner in a Singapore polyclinic. When an AI extracts a medication dose, she will not sign that AI-generated dose unless she can verify it in under three seconds. If the AI provides a value without proving where it came from, she must re-read the entire unstructured dictation to ensure the AI did not hallucinate. In healthcare, a plausible wrong dose costs far more time and risk than a blank box.

MediExtract changes this by shifting the AI deliverable from a "summary" to a deterministic, verifiable fact. The system turns one dictated note into four schema-valid fields (medication, dose, frequency, allergy). Crucially, each field carries the verbatim text span it was extracted from. If the system cannot mathematically ground the value in a direct quote, it blanks the field. By anchoring values to quotes, MediExtract respects the three-second cognitive threshold, allowing rapid visual pattern recognition to replace manual transcript auditing.

2. Architectural Trade-Offs and Performance Tuning

Our architecture deliberately avoids maximum AI usage in favor of targeted AI integration. We utilized a rented foundation model (Gemini 2.5 Flash via OpenRouter) prompted for strict structured output, rather than deploying complex agentic loops or Retrieval-Augmented Generation (RAG).

This decision was driven by empirical measurement and OWASP threat models. RAG was rejected because there is no external knowledge to retrieve; the answer is always inside the note being processed. Furthermore, avoiding multi-agent tool use structurally eliminated OWASP LLM03 (Excessive Agency).

By limiting the LLM to probabilistic reasoning and owning the deterministic verification in Python, we achieved a pooled recall of 0.7297 on our sealed gold-v2 corpus (n=67). We proved the value of this AI integration by re-scoring a baseline (a 14,689-name RxNorm regex gazetteer) against the exact same answer key, which yielded only 0.4324. The paired exact McNemar test confirmed our AI approach separated from the baseline with 52 flips right against 8 wrong ($p = 5.2e-09$).

Financially, this architecture prioritizes clinical time over compute costs. While our cost per note is fractions of a cent (\$0.000617), the explicit cash price of forcing the model to output evidence quotes is approximately \$1.90 per physician-year. At a clinician rate of SGD 100/hour, token prices are irrelevant compared to accuracy; a 5-percentage-point accuracy gain is worth 561 times a token-price saving.

3. Metrics Performance Critique: Honest Limitations

A core requirement of this project was demonstrating reasoning depth by acknowledging rough edges. We recorded a silent-failure rate of 0.2517. Rather than obfuscating this number, we forensically localized the defect.

The data revealed that 31 of these 36 silent failures (86%) stem from a single structural defect: the single-medication slot. In multi-drug notes, the system is forced to select only one drug. 17 failures were direct drug-selection misses, and 14 were the cascading failures of that specific drug's dose and schedule.

Crucially, zero failures are owned by the LLM layer, zero are owned by the gate layer, and all 36 returned values were present in the source note (meaning there is zero hallucination in the set). When the drug choice agrees, our recall is 0.990; when it does not, recall falls to 0.196. This proves the 0.2517 failure rate is a rigid limitation of our v1 JSON schema, not a generative defect of the language model.

4. Evaluations Critique and Difficulties Surpassed

The most significant difficulties we surpassed involved confronting our own flawed hypotheses regarding safety guardrails. In evaluating the system, we specified three distinct mitigation controls to improve performance. We formally measured them, and based on the empirical data, we refused to ship all three.

A. The ALL-CAPS Header Discovery & Recall Gates During testing, we discovered that stripping ALL-CAPS section headers dropped allergy recall from 0.900 to 0.650. We initially interpreted this as the model relying on formatting as a crutch. To mitigate this, we specified a proximity gate and an allergy recall check (S-16).

However, measurement proved both controls were effectively clinical placebos. The proximity gate caught 0 of 36 silent failures at every window size, and broadening its parameters merely destroyed 9 correct fields. The S-16 recall check scored a sensitivity of 0 of 4.

By auditing the failures, we discovered that 201 ALL-CAPS headers survive in the intact notes against only 1 in the stripped set. The header was not a formatting crutch; it was the actual clinical signal differentiating an allergy from a medication. The 0.650 recall is a ceiling set by the input, not a defect to gate away. We formally refused to ship both gates, demonstrating that in clinical AI, acknowledging a documented limitation is safer than shipping security theater.

B. The LLM-as-a-Judge Refusal We built an LLM judge (S-14) using a different model family to evaluate extractions. We set a pre-implementation threshold requiring 0.9 agreement with human labels on both correct and incorrect classes. The judge scored 0.9467 on correct calls, but a dismal 0.2093 on wrong calls. Ultimately, the real judge scored 0.8284, failing to beat a baseline "rubber stamp" that blindly guessed "correct" every time (0.8396). We refused to adopt the judge, ensuring that deterministic hand-labelling continues to stand behind every prompt change.

C. Red-Teaming Prompt Injections Because dictated notes are essentially untrusted input, Prompt Injection (LLM01) is our highest security risk. We red-teamed the pipeline with malicious vectors. While the foundational LLM failed and occasionally obeyed injected commands, our deterministic input scan forced abstentions on 16 attack vectors with 0 false positives. The generative layer failed, but the software perimeter held.

5. The Future Path: Schema Expansion

The future trajectory of MediExtract is explicitly defined by our metrics critique. Because 86% of our silent-failure rate is owned by the single-medication slot, the immediate next step is migrating from a single JSON object to a multi-item medication array.

Preliminary spikes indicate the LLM provider accepts the 12-item schema array effortlessly. However, deploying it requires re-labeling all 67 cases to create a gold-v3 corpus with multi-drug containment labels. For this End-of-Course submission, we prioritized the paired statistical comparability of our baseline over a rushed deployment of the array. Implementing the multi-drug schema is projected to save an additional ~\$3,600 per physician-year, making it the definitive next phase for commercial viability.

Conclusion

MediExtract demonstrates that the successful enterprise deployment of Large Language Models requires limiting their agency. By bounding a probabilistic model within deterministic Python verification gates, we successfully transformed generative AI from a liability that requires constant human auditing into a secure, sub-3-second verification engine for clinical data.